

annually to reach his goal?

41. Jo and Kim want to set aside some money for a down payment on a new car. They have 6 years to let the money grow. If they want to make a \$15,000 down payment on the car, how much should they deposit now in an account that earns 4.36% interest compounded monthly?
42. A newspaper's business section runs an article about savings at various banks in the city. They find six that offer accounts that offer compound interest.
Bank A offers 3.76% compounded daily.
Bank B offers 3.85% compounded annually.
Bank C offers 3.77% compounded weekly.
Bank D offers 3.74% compounded daily.
Bank E offers 3.81% compounded semi-annually.
To earn the most interest on a deposit, which bank should a person choose?
43. Paola reads the newspaper article from exercise 32. She really wants to know how different they are in terms of dollars, not effective annual yield. She decides to compute the future value for accounts at each bank based on a principal of \$100,000 that are allowed to grow for 20 years. What is the difference in the future values of the account with the highest effective annual yield, and the account with the second highest effective annual yield?
44. Paola reads the newspaper article from exercise 32. She really wants to know how different they are in terms of dollars, not effective annual yield. She decides to compute the future value for accounts at each bank based on a principal of \$100,000 that are allowed to grow for 20 years. What is the difference in the future values of the account with the highest effective annual yield, and the account with the lowest effective annual yield?
45. Jesse and Lila need to decide if they want to deposit money this year. If they do, they can deposit \$17,400 and allow the money to grow for 35 years. However, they could wait 12 years before making the deposit. At that time, they'd be able to collect \$31,700 but the money would only grow for 23 years. Their account earns 4.63% interest compounded monthly. Which plan will result in the most money, depositing \$17,400 now or depositing \$31,700 in 12 years?
46. Veronica and Jose are debating if they should deposit \$15,000 now in an account or if they should wait 10 years and deposit \$25,000. If they deposit money now, the money will grow for 35 years. If they wait 10 years, it will grow for 25 years. Their account earns 5.25% interest compounded weekly. Which plan will result in the most money, depositing \$15,000 now or depositing \$25,000 in 10 years?

6.5 Making a Personal Budget



Figure 6.7 Calculating a budget is important to your financial health. (credit: "Budget planning concept on white desk" by Marco Verch Professional Photographer/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Create a personal budget with the categories of expenses and income.
2. Apply general guidelines for a budget.

"That doesn't fit in the budget."

"We didn't budget for that."

"We need to figure out our budget and stick to it."

A **budget** is an outline of how money and resources should be spent. Companies have them, individuals have them, your college has one. But do you have one?

Creating a realistic budget is an important step in careful stewardship of your financial health. Designing your budget will help understand the financial priorities you have, and the constraints on your life choices. You want to have enough income to pay not only for the necessities, but also for things that represent your wants, like trips or dinner out. You also may want to save money for large purchases or retirement. You do not want to just get by, and you do not want the problems associated with overdue balances, rising debt, and possibly losing something you have worked hard to obtain.

While creating a budget may seem intimidating at first, coming up with your basic budget outline is the hardest part. Over time, you will adjust not only the numbers, but the categories.

Creating a Budget

You should view creating a **budget** as a financial tool that will help you achieve your long-term goals. A budget is an estimation of income and expenses over some period of time. You will be able to track your progress, which will help you to prepare for the future by making smart investment decisions.

There are several budget-creating tools available, such as the apps Good budget and Mint, and Google Sheets. Getting started, though, begins well before you find an app. The following are steps that can be used to create your monthly budget.

1. **Track your income and expenses** Review your income and expenses for the past 6 months to a year. This will give you an idea of your current habits.
2. **Set your income baseline** Determine all the sources of income you will have. This income may come from paychecks, investments, or freelance work. It even includes child support and gifts. Be sure to use income after taxes. This allows you to determine your maximum expenditures per month.

 *For income that is not steady, such as gig work or freelance work, use the previous 6 to 12 months of income to find an average income from that gig or freelance work. Use this average in the budgeting process.*

3. **Determine your expenses** Review your bills from the past 6 months. You should include mortgage payments or rent, insurance, car payments, utilities, groceries, transportation expenses, personal care, entertainment, and savings. Using your credit card statements and bank statements will help you determine these amounts. Be aware that some of the expenses will not change over time. These are referred to as **fixed expenses**, like rent, car payments, insurance, internet service, and the like. Other expenses may vary widely from month to month and are appropriately called **variable expenses**, and include such expenses as gasoline, groceries.

 *Some expenses are yearly, such as insurance or property taxes. Other expenses may be quarterly (four times per year) or semiannual (twice per year). To budget for such bills by month, divide the bill total by the number of months the bill covers.*

4. **Categorize your expenses** These categories may be housing, transportation, or food, for broad categories, or may get more specific, where you categorize car payments, car insurance, and gasoline separately. The categories are your choices. Be sure to account for the cost of maintaining a vehicle or home. The more specific you are, the better you'll understand your spending needs and habits.
5. **Total your monthly income and monthly expenses and compare** These values should be compared. If your expenses are higher than your income, then adjustments have to be made. Decisions of what to do with any extra income is part of the planning process also.
6. **Make plans for unplanned expenses** Ask anyone, an unexpected car repair can ruin a carefully crafted budget. Have a plan for how you can be ready for these random expenses. This often means creating a cushion in your

budget.

7. **Use your budget to make decisions and adjust for any changes** Your budget is a changeable document. Add to it when you wish, refer to it when special purchases are to be made. Keeping your budget up to date helps accommodate changes in income and expenses.

▶ VIDEO

[Creating a Budget \(https://openstax.org/r/creating_budget\)](https://openstax.org/r/creating_budget)

In this section, we will focus on income and expenses. One of the easiest ways to manage a budget is to create a table, with one column containing income sources, another with income values, a third with expense categories, and a last containing expenses. An example is shown in [Table 6.1](#).

Income Source	Amount	Expense	Amount
Full-time job	\$3,565	Rent	\$975
Uber	\$185	Car Payment	\$355
TOTAL	\$3,750	Student Loan	\$418
		Electric	\$76
		Food	\$400
		Gasoline	\$250
		Car Insurance	\$165
		Clothing	\$100
		Entertainment	\$100
		TOTAL	\$2,839

Table 6.1 Table with Budget

?? WHO KNEW?

Gross Pay and Take Home Pay

If you've ever had a paycheck, you know that taxes are taken out of your pay before you get your check. This amount of money varies from state to state, and sometimes even city to city. For a person making \$50,000 per year gross salary in Salt Lake City, Utah, take home pay is about 75.6% of gross salary. In Detroit, Michigan, take home pay is about 74.5% of gross salary. Lakeland, Florida, take home pay is about 80.5% of gross salary. These also change based on how much a person earns! Before choosing a place to live, it makes sense to determine how much deductions from pay will impact your income.

EXAMPLE 6.49

Creating a Budget

Heather has graduated college and currently works as a nurse for a rural medical group. Her net monthly income from that job is \$3,765.40. She also works part-time on the weekends, earning another \$672.00 per month. Her monthly expenses are rent at \$1,050, car payments at \$489, student loan payments at \$728, car insurance at \$139, utilities at \$130, clothing at \$150, entertainment (going out with friends, Netflix, Amazon Prime, movies) at \$300, credit card debt at

\$200, food at \$360, and gasoline at \$275. Create her budget in a table, compare the total income to total expenses, and determine how much excess income per month she has or how much she falls short by each month.

 Solution

Step 1: To begin, we create the table with appropriate headings.

Income Source	Amount	Expense	Amount

Step 2: Her income categories are her nursing job, with \$3,765.40 per month, and her part-time job, with \$672.00 per month. Entering these into the table, we have the following.

Income Source	Amount	Expense	Amount
Nursing	\$3,765.40		
Part-time	\$672.00		

Step 3: Her monthly expenses are listed above. Entering the categories and the amount for each of those expenses, the table is now

Income Source	Amount	Expense	Amount
Nursing	\$3,765.40	Rent	\$1,050
Part-time	\$672.00	Car Payment	\$489
		Student Loan	\$728
		Car Insurance	\$139
		Utilities	\$130
		Clothing	\$150
		Entertainment	\$300
		Credit Card	\$200
		Food	\$400
		Gasoline	\$250

Step 4: Totaling the income and expenses, we see that her total income is \$4,437.40 per month, and her total expenses are \$3,836 per month. Comparing these, we see that Heather has \$601.40 in excess income per month. This provides a cushion in her budget.

> YOUR TURN 6.49

1. Mateo works as a union electrician in a suburban area. Monthly, his take home pay is \$3,375. He sometimes does small side jobs for family or friends, and averages about \$300 per month from these little jobs. His monthly expenses are his mortgage at \$986.78, truck payments at \$589.00, truck insurance at \$312, utilities at \$167, clothing at \$150, entertainment at \$400, credit card debt at \$325, food at \$470, and gasoline at \$375. Create Mateo's budget in a table, compare the total income to total expenses, and determine how much excess income per month he has or how much he falls short by each month.

EXAMPLE 6.50

Creating a Budget

Carol is working in a dental lab, creating dentures and bridges. Monthly her take home pay is \$2,816 (based on \$22 per hour minus payroll taxes). She also receives \$320 per month in child support for her one daughter. Her monthly expenses are rent at \$700, car payments at \$229, student loan payments at \$250, car insurance at \$119, health insurance at \$225, utilities at \$80, clothing at \$75, entertainment at \$200, food at \$275, and gasoline at \$275. Create Carol's budget in a table, compare the total income to total expenses, and determine how much excess income per month she has or how much she falls short by each month.

✓ Solution

Step 1: To begin, we create the table with appropriate headings.

Income Source	Amount	Expense	Amount

Step 2: Her income categories are from work, \$2,816, and child support, \$320, per month. Entering these into the table, we have the following.

Income Source	Amount	Expense	Amount
Job	\$2,816.00		
Child support	\$320.00		

Step 3: Her monthly expenses are listed above. Entering the categories and the amount for each of those expenses, the table is now

Income Source	Amount	Expense	Amount
Job	\$2,816.00	Rent	\$700
Child support	\$320.00	Car Payment	\$229
		Student Loan	\$250
		Car Insurance	\$119
		Utilities	\$80

Income Source	Amount	Expense	Amount
		Health insurance	\$225
		Clothing	\$75
		Entertainment	\$200
		Food	\$275
		Gasoline	\$275

Step 4: Totaling the income and expenses, we see that her total income is \$3,136.00 per month, and her total expenses are \$2,428.00 per month. Comparing these, we see that Carol has \$708.00 in excess income per month. This is the cushion in her budget.

> YOUR TURN 6.50

1. Maddy works as a mechanical engineer, making \$6,093.75 monthly after payroll taxes. Her monthly expenses are her mortgage at \$1,452.89, car payments at \$627.38, car insurance at \$179.00, health insurance at \$265.00, utilities at \$320, clothing at \$150, entertainment at \$400, credit card debt at \$450, food at \$370, and gasoline at \$175. Create Maddy's budget in a table, compare the total income to total expenses, and determine how much excess income per month she has or how much she falls short by each month.

Using the budget process, we can make decisions on adding expenses to the budget. To do so, check the cushion of the budget to see if there is room in the budget for the new expense.

EXAMPLE 6.51

Adding to an Existing Budget

In the example above, Carol had excess income of \$708.00. She looks up the cost of before-school care for her daughter. She finds that, monthly, the cost would be \$252.00 per month. Is this an affordable program for Carol? Add this expense to her budget table.

✓ Solution

She can afford this, as the cost for the before school program is \$252.00 and she had extra income of \$708.00. Adding this to her budget, her budget table is now

Income Source	Amount	Expense	Amount
Job	\$2,816.00	Rent	\$700
Child support	\$320.00	Car Payment	\$229
		Student Loan	\$250
		Car Insurance	\$119
		Utilities	\$80

Income Source	Amount	Expense	Amount
		Health insurance	\$225
		Clothing	\$75
		Entertainment	\$200
		Food	\$275
		Gasoline	\$275
		Before-school care	\$252

Now, she has \$456.00 in excess income per month.

> YOUR TURN 6.51

1. Recall Heather's budget from Example 6.49. She decides she wants to buy her own home, which would increase her expenses. Instead of \$1,050.00 in rent, she would pay \$1,240.00 for her mortgage. Her utilities costs would increase to \$295.00 per month. Add these to Heather's budget to determine if the changes are affordable.

The 50-30-20 Budget Philosophy

It isn't clear, obvious, or easy to decide how much of your income to allocate to various categories of expenses. Many people pay their bills and then consider all the leftover money to be spending money. However, when developing your own budget, you may want to follow the **50-30-20 budget philosophy**, which provides a basic guideline for how your income could be allocated. Fifty percent of your budget is allotted to your needs, 30% of your budget is allotted to pay for your wants, and 20% of your budget is allotted for savings and debt service (paying off your debts).

Knowing what expenses are necessary and what expenses are wants is important, since wants and needs are often confused. The following are **necessary expenses** that represent basic living requirements and debt services. This list isn't complete: mortgage/rent, utilities, car, car insurance, health care, groceries, gasoline, child care (for working parents), and minimum debt payments. The 50-30-20 budget philosophy suggests that 50%, or half, your income go to these necessities.

Wants, though, are things you could live without but still wish to have, such as Amazon Prime, restaurant dinners, coffee from Starbucks, vacation trips, and hobby costs. Even a gym membership or that new laptop are wants. Creating the room to afford these wants is important to our mental health. Not budgeting for things we want will negatively impact our quality of life.

The remaining 20% should be set aside, either in retirement funds, stocks, other investments, an emergency fund (recommendations are that an emergency fund have 3 months of income), and perhaps extra spent to pay down debt. This 20% is very useful for addressing those unexpected costs, such as repairs or replacement of items that no longer work. Without budgeting this cushion, any expense that is a surprise can cause us to miss necessary payments.

The list of necessary expenses was not complete. There are other expenses that could be included.

Necessary Expenses and Expenses that are Wants

For some people, an expense will be necessary while the same expense for someone else will be a want. A good example of this is internet service. Many people consider internet service as a need, especially those who work from home or who are not able to leave their homes. One could also call internet service a need if they have children in school. For others, internet service is a want. If a person's job doesn't require them to be online, if they are not in school, if they do not have kids, then internet service can be dropped. There are public options for internet service. One could even use their phone as a hot spot.

Cars often fall into the category of need, but could also fall into the want category, depending on where and how you

live. Bikes, public transportation, and walking are all options that could replace a car. This would then remove the cost of gasoline and car insurance.

Another consideration when deciding if an item on your budget is a need or a want is about your choices and priorities. A car is a need for many. But the need for a car is not the same as the need for a specific car. If you choose to buy a car with payments that exceed your budgeted amount for the car, then that car is a want. The amount you exceed the budget now belongs in the want category.

The same can be said for housing. If you want an apartment that costs \$1,250 per month, but your budget only allows for an apartment that costs \$900, then \$350 of the rent is a want.

The point of that is to carefully consider if an expense is a need as opposed to a want.

When your expenses exceed your income, you may want to change how you budget your income to line up with these guidelines. This may mean cutting back, finding less-expensive living arrangements, finding a less-expensive (and more fuel-efficient) car, or sacrificing some specialty groceries. Using these guidelines keeps your financial life manageable.

Better still, they can guide you as you begin your life after graduation.

EXAMPLE 6.52

Evaluate a Budget Using 50-30-20 model

In the example above, after Carol added before school care for her daughter to the budget, her budget was as shown below. Evaluate Carol's budget using the 50-30-20 budget philosophy.

Income Source	Amount	Expense	Amount
Job	\$2,816.00	Rent	\$700
Child support	\$320.00	Car Payment	\$229
		Student Loan	\$250
		Car Insurance	\$119
		Utilities	\$80
		Health insurance	\$225
		Clothing	\$75
		Entertainment	\$200
		Food	\$275
		Gasoline	\$275
		Before-school care	\$252

Solution

Carol's total income is \$3,136.00. Applying the 50-30-20 budget philosophy to this income requires the calculation of each of those percentages.

For the necessities, Carol should budget 50% of her income, or $0.5 \times \$3,136.00 = \$1,568.00$.

For her wants, she should budget 30% of her income, or $0.3 \times \$3,136.00 = \940.80 .

For savings and extra debt service, she should budget 20% of her income, or $0.2 \times \$3,136.00 = \627.20 .

In her budget, her necessities include all expenses except for entertainment. These expenses total \$2,480, which exceeds the suggested budget amount of \$1568.00. To follow the guidelines, Carol would have to cut back on these necessities.

For her wants, she spends \$200.00 on entertainment, which is well below the suggested budget amount of \$940.80. If she modifies how much she spends on needs, she may be able to increase the spending on her wants.

Her excess income is \$456.00, which is below what she should be saving and using to pay down extra debt. If she does adjust how much she spends on needs, she could increase the amount for savings.

> YOUR TURN 6.52

1. Recall Heather's budget from Example 6.49, before she thought of moving. That budget is below. Evaluate Heather's budget using the 50-30-20 budget philosophy.

Income Source	Amount	Expense	Amount
Nursing	\$3,765.40	Rent	\$1,050
Part-time	\$672.00	Car Payment	\$489
		Student Loan	\$728
		Car Insurance	\$139
		Utilities	\$130
		Clothing	\$150
		Entertainment	\$300
		Credit Card	\$200
		Food	\$400
		Gasoline	\$250

EXAMPLE 6.53

Creating a Budget Based on the 50-30-20 Budget Philosophy

Carmen is about to graduate and has been offered a job at a bank as a data scientist. She estimates her monthly take home pay to be \$5,662.50. Apply the 50-30-20 philosophy to that monthly income. How should Carmen use this information?

✓ Solution

Step 1. To apply the 50-30-20 budget philosophy to Carmen's income, she needs to calculate 50%, 30%, and 20% of her income. Fifty percent of her income is $0.5 \times \$5,662.50 = \$2,831.25$. Thirty percent of her income is $0.3 \times \$5,662.50 = \$1,698.75$. Twenty percent of her income is $0.2 \times \$5,662.50 = \$1,132.50$.

Step 2. She would then budget \$2,831.25 for her needs, \$1,698.75 for her wants, and \$1,132.50 for savings and debt service.

Step 3. When choosing where to live, what to eat, and what to drive, she should make choices that keep those costs, combined with her debt service costs, gasoline, and utilities, below \$2,831.25. This means she will have to make

decisions about what her priorities are.

Step 4. She should then figure out what she wants to do with her money, and stay within the limits, that is, keep those costs below \$1,698.75.

Step 5. Finally, she can begin building her savings with the remaining \$1,132.50.

YOUR TURN 6.53

1. Elijah has finished an apprenticeship and is about to start his first job as an HVAC (heating, ventilation, and air conditioning) tech. He estimates that his net monthly income will be \$3,263.44. Apply the 50-30-20 budget philosophy to his income to set guidelines for Elijah's budget. How should Elijah use this information?

EXAMPLE 6.54

Using the 50-30-20 Budget Philosophy to Analyze Affordability

Steve is thinking of moving out of his family's home. He currently works at a full-time job making \$18 per hour, which will give him, approximately, a net annual income of \$29,180 (working 40 hours per week for 52 weeks per year). He has student debt that he pays off at \$218.00 per month, and already owns a car that he pays \$162.00 per month for.

1. Apply the 50-30-20 budget philosophy to Steve's income.
2. If he follows the budget, how much does he have, after paying his car payment and student loan, to spend on necessities.
3. If he follows the budget, how much will he set aside for wants? For savings?
4. Discuss the affordability of moving out, based on Steve's budget.

Solution

Before the 50-30-20 philosophy can be applied, Steve's monthly income needs to be determined. This is found by dividing his annual income by 12. This gives $\$29,180/12 = \$2,431.67$. This will be used for his monthly budget.

1. To apply the 50-30-20 philosophy to Steve's income, find 50%, 30%, and 20% of his monthly income.
Needs (50%): $50\% \text{ of his income is } 0.50 \times \$2,431.67 = \$1,215.83$.
Wants (30%): $0.30 \times \$2,431.67 = \729.50
Savings (20%): $0.20 \times \$2,431.67 = \486.33
2. The total for Steve's needs is \$1,215.83. From this, he already pays \$218.00 for his student loans, and \$162.00 for his car payment. Together that is \$380.00. Subtracting from the amount he should budget for his needs, he can spend \$835.83 on other needs.
3. Steve budgeted \$729.50 for wants, and \$486.33 for savings and other debt servicing.
4. Steve will have other needs to pay for, including rent, utilities, food, health care, gasoline, and car insurance. It is difficult to imagine Steve being able to afford to move out, unless he reallocates money that he would want to save, or use for entertainment and other wants, or takes on another job. Even if Steve uses all the money that the 50-30-20 budget sets aside for savings, he still only has \$1,322.16 to spend on those necessities. It does not appear he can afford to move out.

YOUR TURN 6.54

Fran wants to take a new job but will have to move to an area with a higher cost of living. With her current income, she can use the 50-30-20 budget philosophy. The new job will have a net pay of \$43,700 annually. She will still have to pay her car payment of \$295.00, her student loans that cost \$264.00 per month, and her outstanding credit card debt, on which she pays \$200 per month.

1. Apply the 50-30-20 budget philosophy to Fran's new income.
2. If she follows the budget, how much does she have, after paying her credit card debt, car payment and student loan, to spend on necessities.
3. If she follows the budget, how much will she set aside for wants? For savings?

4. Discuss the affordability of changing jobs and moving, based on Fran's budget.

 VIDEO

50-30-20 Budget Philosophy (https://openstax.org/r/50-30-20_budgeting_rule)

Check Your Understanding

26. What is a budget?
27. What are necessary expenses?
28. David gathers his paystubs and bills from the past 6 months. His income, after taxes, is \$3,450 per month. His rent, utilities included, is \$925. His car payments are \$178.54 per month, his car insurance is \$129.49 per month, his credit cards cost him \$117.00 per month, he spends \$195 per month on gas, his food costs are \$290 per month. He also spends \$21.99 on Amazon prime, \$49.99 on his internet bill, and \$400 per month going out. Create David's monthly budget, including totals, based on that information.
29. Using David's Budget from Exercise 28, how much income does he have per month after accounting for his expenses?
30. Apply the 50-30-20 budget philosophy to David's budget.
31. Evaluate David's budget with respect to the 50-30-20 budget philosophy.



SECTION 6.5 EXERCISES

In the following exercises, categorize each expense as a necessary expense or an expense that is a want.

1. Rent
2. Dinner at a restaurant.
3. Car payment
4. New game system
5. Gym membership
6. Electric bill
7. Heating bill
8. Phone bill
9. Netflix
10. Student Loan Payment
11. Explain how a necessary expense for one person could be a want expense for another person.
12. Explain how a necessary expense may be partly a necessary expense and partially a want expense.

In the following exercises, create the budget, including totals and how much the income exceeds or falls short of the expenses, based on the information given.

13. Per month: paychecks = \$3,680, consulting = \$900, Mortgage = \$1,198.00, Utilities = \$376, Cell phone = \$67.50, Car payments = \$627.85, Car insurance = \$183.50, Student loans = \$833, Food = \$450, Gasoline = \$275, Internet = \$69, Dining out = \$250, Credit cards = \$375, entertainment = \$300
14. Per month: paychecks = \$2,750, child support = \$500, Mortgage = \$945.50, Utilities = \$195, Cell phone = \$37.50, Car payments = \$298.23, Car insurance = \$163.50, Student loans = \$438, Food = \$250, Gasoline = \$175, Internet = \$49, Netflix = \$15, After school care = \$711, Credit cards = \$150, entertainment = \$150
15. Per month: paychecks = \$4,385, Rent = \$1095, Utilities = \$165, Cell phone = \$67.50, Car payments = \$467.35, Car insurance = \$243.75, Student loans = \$1,150, Food = \$325, Gasoline = \$260, Internet = \$99, Netflix = \$15, Amazon = \$23, Gym membership = \$49, entertainment = \$650
16. Per month: paychecks = \$3,460, Gig job = \$173, Rent = \$895, Utilities = \$165, Car payments = \$195.80, Car insurance = \$123.30, Food = \$265, Gasoline = \$185, Internet = \$39, Hulu = \$15, Amazon = \$23, Credit cards = \$97.60, Entertainment = \$600

In the following exercises, determine the amount of money that should be allocated to each of the three categories of the 50-30-20 budget philosophy guidelines.

17. Referring to Exercise 13: Monthly income = \$4,580.00
18. Referring to Exercise 14: Monthly income = \$3,250.00

19. Referring to Exercise 15: Monthly income = \$4,385.00

20. Referring to Exercise 16: Monthly income = \$3,633.00

In the following exercises, evaluate the given budget with respect to the 50-30-20 budget philosophy guidelines.

21. The budget and 50-30-20 rule from exercises 13 and 17.

22. The budget and 50-30-20 rule from exercises 14 and 18.

23. The budget and 50-30-20 rule from exercises 15 and 19.

24. The budget and 50-30-20 rule from exercises 16 and 20.

For the following exercises, Kiera and Logan sit down to make their budget. Kiera works full time as a mental health counselor and sells kids toys on her own. Logan works as a branch manager at a local bank and works part-time at the nearby bar. They collect their financial document to work out their budget. Kiera's paychecks from her job as a mental health counselor, after taxes and per month, total \$3,021. Logan's paychecks from the bank, after taxes and per month, total \$3,827. Kiera's income from toy sales for the last 3 months were \$140, \$87, and \$475. Logan's take-home pay from the bartending job for the last 3 months were \$540, \$310, and \$449.

25. Determine how much income Kiera and Logan have per month.

26. Apply the 50-30-20 budget philosophy to their income.

For the following exercises, Kiera and Logan gather their bills from the last 6 months. Their fixed expenses, with costs, are rent for \$1,350, Kiera's car payment for \$275, Logan's car payment of \$380, student loans (they each have students loans) for \$934, car insurance for \$289, internet service for \$39, Netflix for \$15, Amazon Prime for \$24, gym membership for \$99, and cell phones for \$250. The variable cost expenses, and their average costs for the last 6 months, are utilities for \$370, gasoline for \$500, food for \$475, clothing for \$225, and miscellaneous entertainment expenses for \$535. They always pay off their credit card bill and carry no balance.

27. Create their budget, using the income from Exercise 25.

28. Categorize each expense as a need or a want. Find the total for each, along with remaining income.

29. Compare their budget to the guidelines from the 50-30-20 budget from Exercise 27.

30. Determine if Kiera and Logan can afford to buy a new computer, which would cost \$330 per month for the next 6 months.

In the following exercises, [the Federal Paycheck Calculator \(https://openstax.org/r/smartasset\)](https://openstax.org/r/smartasset) was used to estimate monthly take-home pay. The annual salary, before taxes and deductions, is provided. Then, the monthly take-home pay after taxes and deductions is given (which means the monthly take-home pay is not just the annual salary divided by 12!). In each case, apply the 50-30-20 budget philosophy to the monthly take-home income. Note: These are based on living in Indianapolis, Indianapolis, unmarried and with no dependents.

31. Annual salary: \$30,000. Monthly take home: \$1,938

32. Annual salary: \$40,000.00. Monthly take home: \$2,564

33. Annual salary: \$50,000. Monthly take home: \$3,144

34. Annual salary: \$70,000. Monthly take home: \$4,229

35. Annual salary: \$100,000. Monthly take home: \$5,840

36. Annual salary: 150,000. Monthly take home: \$8,506

In the following exercises, [the Federal Paycheck Calculator \(https://openstax.org/r/smartasset\)](https://openstax.org/r/smartasset) was used to estimate monthly take-home pay. The hourly pay, before taxes and deductions, is provided. Then, the monthly take-home pay after taxes and deductions is given (which means the monthly take-home pay is not just the hourly pay times 174 hours!). In each case, apply the 50-30-20 budget philosophy to the monthly take-home income. Note: These are based on living in Tempe, Arizona, unmarried and with no dependents.

37. Hourly pay: \$12.15 (minimum wage in Tempe, Arizona as of September 2022). Monthly take home: \$1,698

38. Hourly pay: \$15.00. Monthly take home: \$2,083

39. Hourly pay: \$17.50. Monthly take home: \$2,421

40. Hourly pay: \$19.75. Monthly take home: \$2,725

41. Hourly pay: \$25.00. Monthly take home: \$3,369

42. Hourly pay: \$35.00. Monthly take home: \$4,547